AFC–10369

Sub. Code
1BCA1C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

First Semester

Computer Application

PROGRAMMING IN C

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a token? Give an example.

2. What are the types of expressions available in C?

3. How will you initialize a matrix?

4. Write any four string handling functions in C.

5. Define function.

6. How will you access the members of a structure?

7. Write any two advantages of pointer.

8. Write a pointer expression for the equation y=((a-b)*c)/(c-d);

9. What is the difference between an ASCII file and binary file?

10. What is the purpose of compiler control directives?

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the operators in C with example.

Or

(b) Explain else-if ladder statement with an example.

12. (a) Write a C program to count the number of characters in the given string without using string handling functions.

Or

(b) Explain 2D arrays with an example.

13. (a) Write a C program to calculate the factorial of a given number using recursion.

Or

(b) Explain with an example how will you initialize a structure.

14. (a) Write a C program to find the biggest of three numbers using pointers.

Or

(b) Write a C program to find the smallest number in an array using pointers.

15. (a) Explain command line argument with a program.

Or

(b) Explain various file opening modes.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain any two loop statements available in C with example.

17. Write a C program to multiply two matrices, if possible.

18. Write a C program to print the pay slip of an employee using array of structures. (Assume your own data).

19. Explain pointers to structure with an example.

20. Explain the steps to open and close a file with example.

AFC–10370

Sub. Code
1BCA2C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Second Semester

Computer Application

PROGRAMMING IN C++

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Encapsulation.

2. What is the use of break statement?

3. What is the difference between static member function and non static member function?

4. What is the purpose of Destructor?

5. What is Direct Base class and Indirect Base class?

6. Write any Four C++ stream classes.

7. What is a Binary File?

8. How will you open a file?

9. What is a class Template?

10. What is the difference between an Exception and an Error?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the operators in C++ with example.

Or

(b) Explain switch statement with an example.

12. (a) Explain Dynamic constructor with an example.

Or

(b) Explain the characteristics of a Constructor.

13. (a) Explain formatted console I/O operations.

Or

(b) Explain Pointer to derived class with an example.

14. (a) Explain File pointers and their manipulators with example.

Or

(b) Explain Command-Line argument with a program.

15. (a) Explain the exception handling model with an example.

Or

(b) Explain exceptions in operator overloaded functions with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a C++ program to print the first 20 prime numbers using while loop.

17. Explain friend function with a program.

18. Explain virtual base class with a program.

19. Write a C++ program to store the students details in a file using write( ).

20. Write a C++ program to find the sum two numbers (i.e, int, float and char types) using function.

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AFC–10371

Sub. Code
1BCA3C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Third Semester

Computer Application

DATABASE MANAGEMENT SYSTEMS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by Data Abstraction?

2. What is a Relationship set? Give an example.

3. What is meant by Temporal data?

4. What is the purpose of Normalization?

5. What do you mean by Client-Server computing?

6. Define Transaction.

7. What is meant by Data Integrity?

8. What are the advantages of View?

9. Write any Four DDL commands.

10. What is a package?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the Database Languages.

Or

(b) Explain E – R diagrams with an example.

12. (a) Explain First Normal form with an example.

Or

(b) Explain the Multivalued Dependencies with an example.

13. (a) Explain the Transaction-Server process structure.

Or

(b) Explain the various Network types.

14. (a) Explain with an example how will you create a Table.

Or

(b) How will you create and delete a User? Explain with an example.

15. (a) How will you create a function and execute a function? Explain.

Or

(b) Explain the various types of Triggers.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain :

(a) Purpose of Database Systems

(b) E-R Design issues.

17. Explain :

    (a) 3NF

    (b) BCNF.

18. Explain the Distributed Data storage.

19. What is Synonym? Explain the creating and Renaming a Synonym with an example.

20. Write a procedure to print the multiplication Table.

AFC–10372

Sub. Code
1BCA4C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Fourth Semester

Computer Application

JAVA PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the basic concepts of Object Oriented Programming?

2. What are the different types of Java statement?

3. What do you mean by type conversion?

4. What do you mean by associativity of arithmetic operator? Give one example.

5. What is the difference between Method overloading and Method overriding?

6. What is the purpose of keyword final?

7. What are the advantages of Thread?

8. Define Exception.

9. What is an Applet?

10. Write the syntax of the method which is used to draw polygon.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Explain the applications of OOP.

Or

(b) Explain the java features.

12. (a) Explain the switch statement with an example.

Or

(b) Write a Java program to find whether the given number is prime number or not a Prime number.

13. (a) Explain static method with a program.

Or

(b) Write a Java program to find the biggest number in an array.

14. (a) Define Package. How will you add classes to a package?

Or

(b) Explain the exception handling in Java with an example.

15. (a) Explain the Applet life cycle.

Or

(b) How will you pass parameters to an Applet? Explain with an example.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain command line arguments with a program.

17. Write a Java program to solve the Quadratic equations for all cases.

18. How will you implement multiple inheritance using interface? Explain with a program.

19. Explain with a program, how will you create the thread by implementing runnable interface.

20. Write a Java program to draw five concentric circles.

AFC–10373

Sub. Code
1BCA5C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Application

VISUAL PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is an event procedure?

2. Write the syntax of For...Next loop.

3. Write the important properties of text box control.

4. List the uses of combo box control.

5. What is the purpose of current X and current Y property?

6. What do you know about tree view control?

7. Write any two differences between MDI and SDI.

8. What is snapshot type record set?

9. Define the term ADO.

10. What is the use of Data grid control?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain do-loop with an example.

Or

(b) Explain subroutines with an example.

12. (a) Explain scroll bar control with an example.

Or

(b) How to build a dynamic form at run time? Explain.

13. (a) Write a program to draw the basic geometric shapes.

Or

(b) Write a program to move a picture.

14. (a) How to format the text using Rich text box control? Explain.

Or

(b) Explain about list view control with example.

15. (a) Write a VB program to access the student database using ADO data control.

Or

(b) How to validate the data in a database?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write a VB program to design a simple calculator.

17. Explain about file controls with example.

18. Write a VB program to display the pictures in random order.

19. Explain about MSFlex Grid control with example.

20. Describe about ADO objects

AFC–10374

Sub. Code
1BCA5C2

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Application

COMPUTER SYSTEM ARCHITECTURE AND DESIGN

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Convert AFb3 in to binary and octal numbers.

2. What is meant by over flow?

3. Write the micro operations for ADD to AC, and LDA.

4. What is an assembler?

5. Write the three types of notations for the equation $A * B + C * D$.

6. What is meant by implied mode of addressing?

7. Draw the hardware for signed-magnitude addition and substraction.

8. Write the purposes of a communication link in I/O interface.

9. Draw the memory hierarchical diagram in a computer system.

10. What is meant by bootstrap loader?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain about the instruction cycle.

Or

(b) Describe the computer registers available in a digital computer.

12. (a) Explain the subroutines with an example.

Or

(b) Explain the Input-Output configuration with a block diagram.

13. (a) Describe the stack organisation.

Or

(b) Explain the general register organisation.

14. (a) Explain the hardware algorithm of multiplication of fixed point numbers.

Or

(b) Explain the interrupt cycle.

15. (a) Explain hit ratio and miss ratio.

Or

(b) Explain the process of writing into cache memory.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain how the basic computer registers are connected to a common bus.

17. Describe the logic and shift operations with example.

18. Describe single accumulator organisation.

19. Describe DMA data transfer with block diagram.

20. What is meant by virtual memory? Explain the memory table for mapping a virtual address.

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AFC–10375

Sub. Code
1BCAE1A

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Application

Elective – WEB DESIGN TECHNOLOGY

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the difference between tag and attributes?

2. What is the significance of <title> tag in HTML?

3. Name any two objects in Javascript and state their purpose.

4. What is the use of break statement?

5. What is a math object?

6. Define the term function.

7. What is an event?

8. List the uses of Dynamic HTML.

9. What is scaling?

10. Write the syntax for IMG element.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain in detail about the <table> tag with example.

Or

(b) Explain with examples the various text formatting attributes.

12. (a) Explain about switch structure in JavaScript with Example.

Or

(b) Write a JavaScript to find the area of a circle.

13. (a) Explain in detail about Recursion with an example.

Or

(b) Describe about string object in JavaScript.

14. (a) Explain about image filters.

Or

(b) Create a webpage with scrolling text.

15. (a) Write the syntax of EMBED element with example.

Or

(b) How will you move shapes?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Design a HTML from for college Admission.

17. Explain the operators in JavaScript.

18. Explain in detail about JavaScript objects.

19. Explain about onfocus and onblur event with example.

20. What are the various filter effects that can be applied to the text? Explain.

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AFC–10376

Sub. Code
1BCAE1B

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Application

Elective: MOBILE COMMUNICATIONS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the use of SHF and VHF?

2. What is meant by Path loss of radio signals?

3. What do you mean by Exposed terminals?

4. What is meant by Handover?

5. What do you mean by Ad-Hoc Network?

6. What is Access Point?

7. What are the limitations of AD HOC networks?

8. What is the purpose of TimeStamp?

9. What is meant by Strong consistency?

10. Define Posting.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the use of Signals in Wireless transmission.

Or

(b) Write short notes on Cellular Systems.

12. (a) Compare the S/T/FCDMA.

Or

(b) Write short notes on Routing.

13. (a) Compare Infrared and Radio Transmission.

Or

(b) Write short notes on HIPERLAN.

14. (a) Explain the functions of Mobile Network layer.

Or

(b) Explain DSR in Mobile Ad-Hoc networks.

15. (a) Explain the functions of HTTP.

Or

(b) Write short notes on WWW.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe various types of Multiplexing schemes.

17. Explain

(a) DECT

(b) DDMA.

18. Explain
    (a) Reference Model
    (b) Addressing.

19. Explain
    (a) Traditional TCP
    (b) Transaction Oriented TCP.

20. Describe the architecture and protocols of WAP.

AFC–10377

Sub. Code
1BCAE2A

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017.

Fifth Semester

Computer Application

***Elective* — COMPUTER GRAPHICS**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define Refresh Rate.

2. Write the equation of a straight line.

3. Define Scaling.

4. Write down the transformation matrix for rotating an object for –90 degree.

5. Define Viewing Window.

6. What is the condition for point clipping?

7. What is the difference between 2D and 3D transformation?

8. What are 3D basic transformations?

9. What is the purpose of user interface?

10. What is the role of Command language in user interface design?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the working of Random Scan displays.

Or

(b) Explain the Circle generation algorithm.

12. (a) Explain the Transformation principles with example.

Or

(b) Explain Composite transformation with an example.

13. (a) Explain window to viewport transformation.

Or

(b) Explain Convex Polygon clipping with an example.

14. (a) Explain 3D rotation with an example.

Or

(b) Explain 3D composite transformation with an example.

15. (a) Explain the user model in User interface design.

Or

(b) Write short notes on Styles of Command language.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write and explain the DDA line drawing algorithm.

17. Describe the basic 2D transformation with example.

18. Explain Line clipping algorithm with an example.

19. Explain the 3D Scaling and Translation transformations with example.

20. Explain :

    (a) Components of user interface

    (b) Feedback.

AFC–10378

Sub. Code
1BCAE2B

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Fifth Semester

Computer Application

Elective — LINUX PROGRAMMING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is open source software?

2. What is inode?

3. What is absolute path? Give an example.

4. Explain cal and who shell commands.

5. What is sort command?

6. What is pipe line?

7. What are the modes of operation in vi editor?

8. What is the purpose of set commands?

9. What is the purpose of break and continue command?

10. What is ftp?

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the advantages of Linux?

Or

(b) Briefly discuss about Linux architecture.

12. (a) What are wild card characters? Explain.

Or

(b) How to change file permission? Explain.

13. (a) Explain about process priorities.

Or

(b) Explain background process.

14. (a) Explain shell procedures with an example.

Or

(b) How to do arithmetic in shell scripts?

15. (a) Explain nested – if with suitable example.

Or

(b) Explain for....loop with an example.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss features of Linux operating system.

17. Explain any five commands used for navigating file system.

18. Explain grip utility and tr filter.

19. Explain the positional parameters and command line arguments in shell programming.

20. Explain the functions of system administrator.

AFC–10379

Sub. Code
1BCA6C1

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Computer Application

DATA MINING AND DATA WAREHOUSING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a Data Warehousing?

2. What is meant by Loading the Data?

3. What is a Process?

4. Why do you need tools to manage a Data Warehouse?

5. What is KDD?

6. What is Data Mining?

7. What do you mean by OLAP?

8. What are the applications of Neural Networks?

9. What is Incremental Rules?

10. What is Item Set?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain the methods for Clean and Transform the Data.

Or

(b) Explain the Data Warehouse architecture.

12. (a) How to estimate the load? Explain.

Or

(b) Explain Tuning queries with example.

13. (a) Explain the Social implications of Data Mining.

Or

(b) Explain the Data Mining issues.

14. (a) Explain the Dimensional modeling.

Or

(b) Explain the Decision Support systems.

15. (a) How will you measure the quality of rules? Explain.

Or

(b) Explain Advanced Association Rule techniques.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the Data Warehouse delivery method.

17. Explain (a) Load Manager (b) System Manager

18. Describe the Data Mining metrics.

19. Explain the statistical perspective on Data Mining.

20. Explain Distributive algorithms for Association rules.

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AFC–10380

Sub. Code
1BCA6C2

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Computer Application

COMPUTER NETWORKS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a Point-to-Point Network?

2. What is meant by Geostationary Satellite?

3. What is the purpose of Error Detection code?

4. What is ALOHA?

5. What are the design issues of Network layer?

6. What is meant by Switching?

7. What do you mean by Upward Multiplexing?

8. Write down the QOS parameters for the Transport layer.

9. What are the advantages of Data compression?

10. What is Plain Text and Cipher Text?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write short notes on ISDN.

Or

(b) Explain any Two guided transmission media.

12. (a) Explain the working of unrestricted Simplex protocol.

Or

(b) Explain the working of Collision free protocol.

13. (a) Explain any one Congestion control algorithm.

Or

(b) Explain the working Link state Routing.

14. (a) Explain how connections are released by Transport layer protocols

Or

(b) Write short notes on Flow Control.

15. (a) Write short note on Network Security.

Or

(b) Explain SNMP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the OSI reference model with a diagram.

17. Explain the working of any Two Sliding window protocols.

18. Explain the Network layer in ATM networks.

19. Explain the various Transport services.

20. Explain :

    (a) Email privacy

    (b) JPEG

AFC–10381

Sub. Code
1BCA6C3

B.C.A. DEGREE EXAMINATION, NOVEMBER 2017

Sixth Semester

Computer Application

SOFTWARE ENGINEERING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the technical parameters of small size projects?

2. What are the quantified and qualified requirements of a software project?

3. Which principle is used in expert judgement method of cost estimation?

4. What is meant by recurrence relation?

5. What is meant by test plan?

6. What is meant by modularization?

7. What is meant b integration testing?

8. Define : validation testing.

9. What are the factors that determine the quality of a software?

10. What are the needs of software quality assurance?

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Explain about the level of technology and level of reliability of a software product.

Or

(b) Explain the planning process of a software project.

12. (a) Explain about the delphi cost estimation technique.

Or

(b) Describe about the estimating software maintenance costs.

13. (a) Explain the cohesion and coupling.

Or

(b) Explain the data flow diagrams.

14. (a) Describe the sources code metrics.

Or

(b) Describe the system testing.

15. (a) Explain about inspections.

Or

(b) Explain about walkthroughs.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the managerial issues of a software project.

17. Describe about the algorithmic cost models.

18. Explain about real time and distributed system design.

19. Explain the managerial aspects of software maintenance.

20. Describe about ISO-9000 quality standards.

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AFC–10493

Sub. Code
1BCASA1/ 1BCA1A1

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer application

Allied – DISCRETE MATHEMATICS

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Define if statement.

2. Construct the truth table for $(P \vee Q) \vee \neg P$.

3. Show that $Q \Rightarrow P \rightarrow Q$.

4. What is an open statement?

5. Define graph.

6. What is cutset?

7. What is Bipartite graph?

8. Define Hamiltonian graph.

9. Define partial ordering of a set.

10. Give an example of a sublattice.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Explain connectives with example and truth table.

Or

(b) Construct the truth table for the following formula

$((P - Q)(\rightarrow Q)) \rightarrow P$

12. (a) Obtain a disjunctive normal form of $P \rightarrow \left((P \rightarrow Q) \wedge \neg(\neg Q \vee \neg P)\right)$.

Or

(b) Explain the rules of generalization and specification.

13. (a) Show that in any graph G, the number of vertices of odd degree is even.

Or

(b) Define

(i) Connected graph

(ii) Component of a graph

14. (a) State and prove Cayley's formula.

Or

(b) Explain Eulerian graph.

15. (a) Write an elaborate note on Partition and covering of a set.

Or

(b) Write in detail about sub lattices.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain Negation and conditional statements.

17. Obtain the PDNF for $(P \wedge Q) \vee (\neg P \wedge R) \vee (Q \wedge R)$.

18. Write in detail about.

    (a) Cutset and cut vertices

    (b) Complete graph.

19. Explain prim's algorithm.

20. State and explain the properties of sublattices.

AFC–10494

Sub. Code
1BCASA2

U.G. DEGREE EXAMINATION, NOVEMBER 2017.

Computer Application

***Allied* — RESOURCE MANAGEMENT TECHNIQUES**

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the role of operations research in decision making?

2. What are the objectives of OR?

3. Define optimal solution.

4. When do you say that an LPP has infinite number of solutions?

5. What is the principle of duality?

6. Define the integer – programming problem.

7. State mathematical form of an assignment problem.

8. Write the method to solve the assignment problem.

9. Write any four applications of transportation problem.

10. Distinguish between balanced and unbalanced transportation problem.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Discuss the scope of OR.

Or

(b) Discuss the various methods of solving an operations research problem.

12. (a) Solve the following LPP by the graphical method :

Maximize $Z = 5x_1 + 4x_2$

Subject to $x_1 + 2x_2 \leq 1$

$x_1 + 2x_2 \geq 3$

$x_1, x_2 \geq 0.$

Or

(b) Use the simplex method to solve the following LPP :

Maximize $Z = 10x_1 + 15x_2 + 20x_3$

Subject to $2x_1 + 4x_2 + 6x_3 \leq 24$

$3x_1 + 9x_2 + 6x_3 \leq 30$

$x_1, x_2 \geq 0.$

13. (a) Apply the dual simplex method to solve :

Minimize $Z = 2x_1 + x_2$

Subject to the constraints $3x_1 + x_2 \geq 3$

$4x_1 + 3x_2 \geq 6$

$x_1 + 2x_2 \geq 3,$

and $x_1, x_2 \geq 0.$

Or

(b) Explain the rules for writing the dual of a primal.

14. (a) Explain Hungarian method for solving an assignment problem.

Or

(b) Solve the following assignment problem :

Machines

		M_1	M_2	M_3	M_4
Jobs	J_1	18	26	17	11
	J_2	13	28	14	26
	J_3	38	19	18	15
	J_4	19	26	24	10

15. (a) Discuss the various steps involved in finding the optimum solution under the MODI method.

Or

(b) Obtain the initial basic feasible solution to the following transportation problem using the least cost method :

	D_1	D_2	D_3	
S_1	1	2	6	7
S_2	8	4	2	12
S_3	3	7	5	11
	10	10	10	

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Discuss the operations research models.

17. Solve using Big-M method :

Maximize $Z = 3x_1 + 2x_2$

Subject to the constraints $2x_1 + x_2 \leq 2$;

$3x_1 + 4x_2 \geq 12$; $x_1, x_2 \geq 0$.

18. Solve the following IPP :

Maximize $Z = 2x_1 + 2x_2$

Subject to the constraints $5x_1 + 3x_2 \leq 8$

$2x_1 + 4x_2 \leq 8$ and are integers.

$x_1, x_2 \geq 0$

19. Solve the following travelling salesman problem :

		Cities			
		A	B	C	D
Cities	A	∞	46	16	40
	B	41	∞	50	40
	C	82	32	∞	60
	D	40	40	36	∞

20. Solve the following transportation problem :

	D_1	D_2	D_3	D_4	Supply
A	6	4	1	5	14
B	8	9	2	7	16
C	4	3	6	2	5
Demand	6	10	15	4	35

AFC–10496

Sub. Code
1BCASA4

U.G. DEGREE EXAMINATION, NOVEMBER 2017

Computer Application

Allied : COST AND MANAGEMENT ACCOUNTING

(CBCS – 2011 onwards)

Time : 3 Hours Maximum : 60 Marks

Section A (10 × 1.5 = 15)

Answer **all** questions.

1. What is a cost sheet?

2. Give example for office overheads.

3. What is Re-order level?

4. Define Material control.

5. What is time rate system?

6. What do you mean by ordinary time wage?

7. What is margin of safety?

8. What is P/V ratio?

9. Write short notes on Zero Base Budget.

10. What is fixed overhead?

Section B (5 × 3 = 15)

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the objectives of costing?

Or

(b) Total cost of sales amounted to Rs. 9,000. The company earns a profit of 10% on sales. Calculate the amount of sales.

12. (a) What are the various types of stock level?

Or

(b) Find out EOQ from the following :

Annual usage 6,000 units, cost of material per unit Rs. 20, cost of placing and receiving one order Rs. 60, annual carrying cost of one unit 10% of inventory.

13. (a) What are the principal system of wage payment?

Or

(b) Calculate the earnings of a worker under Halsey Premium Plan.

Time allowed = 60 hours

Time taken = 50 ours

Rate per hour = Rs. 10.

14. (a) Explain the Marginal cost Equation.

Or

(b) Direct Material Rs. 1,00,000

Direct Wages Rs. 80,000

Fixed Overheads Rs. 30,000

Variable Overheads Rs. 40,000

Sales Rs. 2,70,000

Prepare a marginal cost statement.

15. (a) Explain different types of budget.

Or

(b) Prepare a production budget for three months ending 30.12.2013 for a factory producing four products on the basis of the following information.

Types of product	Estimated stock on 1.10.2013 Units	Estimated sales October-December Units	Desired closing stock on 31.12.2013 Units
A	2,000	10,000	3,000
B	3,000	15,000	5,000
C	4,000	13,000	3,000

Section C (3 × 10 = 30)

Answer any **three** questions.

16. The costing data of Zen Ltd., show the following :

Materials used Rs. 14,00,000

Direct wages Rs. 10,80,000

Factory overhead Rs. 3,24,000

Establishment and General Expenses Rs. 2,24,320

Prepare a statement showing :

(a) Factory cost

(b) Total cost

(c) Percentage of factory overhead that bears to the direct wages.

(d) Percentage of Establishment and General Expenses is related to factory cost.

17. Show the Stores Ledger entries as they would appear when using the LIFO method of pricing of material issues for the following transactions.

Date	Transaction	Units	Price (Rs.)
April 1	Balance in hand b/f	300	2.00
April 2	Purchased	200	2.20
April 4	Issued	150	–
April 6	Purchased	200	2.30
April 11	Issued	150	–
April 19	Issued	200	–
April 22	Purchased	200	2.40
April 27	Issued	150	–

18. A worker takes 9 hours to complete a job on daily wages and 6 hours on a scheme of payment by results. His day rate is 25 paise an hour, the material cost the product is Rs. 4 and the overhead are recovered at 150% of the total direct wages. Calculate the factory cost of the production under the

    (a) Piece work plan,

    (b) Halsey plan and

    (c) Rowan plan.

19. From the following data find out :

    (a) Breakeven Point,

    (b) P/V Ratio,

    (c) Margin of Safety ratio.

    Variable cost Rs. 60,000, Fixed cost Rs. 30,000. Sales Rs. 1,00,000.

20. From the information given below, prepare a flexible budget at 60% and 80% capacities.

	75%
	Rs.
Variable overheads :	
Indirect Materials	7,500
Indirect Labour	22,500

	75%
	Rs.
Semi-variable overheads :	
Electricity (40% fixed)	37,500
Repairs (80% fixed)	3,750
Fixed overheads :	
Salaries	1,00,000
Insurance	5,000
Depreciation	25,000
	2,01,250